BorNEO HackWknd

GROWEASY REPORT

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Introduction

Background of Case Study

Food security has become one of the most pressing strategic issues at both the global and national level. With the world's population continuing to grow and increasing pressure on agricultural land, the ability to produce food efficiently and sustainably is a challenge that can no longer be ignored.

In Indonesia, the agricultural sector plays a vital role not only as the primary food source for over 270 million people, but also as the economic backbone for millions of farming families. However, national agricultural productivity continues to face serious obstacles, including:

- Climate uncertainty affecting planting patterns and harvest outcomes
- Mismatches between crops planted and local rainfall and humidity conditions
- Suboptimal land use due to a lack of soil composition data
- Significant geographical differences across regions that are often overlooked in planting decisions

Traditional farmers typically rely on generational habits or market trends when deciding which crops to plant without considering location-specific scientific data. This could lead to a mismatch between chosen crops and local environmental conditions, ultimately resulting in suboptimal yields, wasted resources, and economic losses.

Meanwhile, individuals who are new to agriculture whether community members looking to contribute to local food security, or simply those who want to try growing their own food for fun or following the trend of urban farming face even greater barriers, such as:

- No agricultural background or foundational knowledge
- Difficulty accessing information that is relevant, structured, and easy to understand
- No awareness of which crops are viable or optimal for their specific region
- No access to agricultural experts or extension workers who can provide guidance

Chosen SDGs

This project directly contributes to two United Nations Sustainable Development Goals:

- **SDG 2: Zero Hunger**
  - Increases agricultural productivity by providing data-driven crop recommendations tailored to specific locations
  - By empowering both farmers and beginners to grow the right crops in the right place, food production becomes more efficient, food security improves thus hunger is reduced
- **SDG 9: Industry, Innovation and Infrastructure**
  - Leverages cutting-edge AI technology to modernize the agricultural decision-making process, making smart farming tools accessible to everyone regardless of technical background
  - Bridges the gap between advanced agricultural data and everyday users by delivering intelligent, location-based recommendations through an intuitive digital platform

Problem Statement

- **Lack of a Simple, Location-Based Crop Recommendation System**
  No free, user-friendly platform currently exists that can automatically detect a user's location, analyze local environmental conditions, and instantly recommend suitable crops with clear explanations. Existing tools are built for professional farmers, require technical manual input, and are largely inaccessible to the general public.
- **No Interactive Support for Agricultural Guidance**
  When farmers or beginners have questions about planting, soil, or crop care, there is no immediate and reliable source to turn to. Agricultural experts are limited in reach, and online resources are scattered leaving most users without the guidance they need.

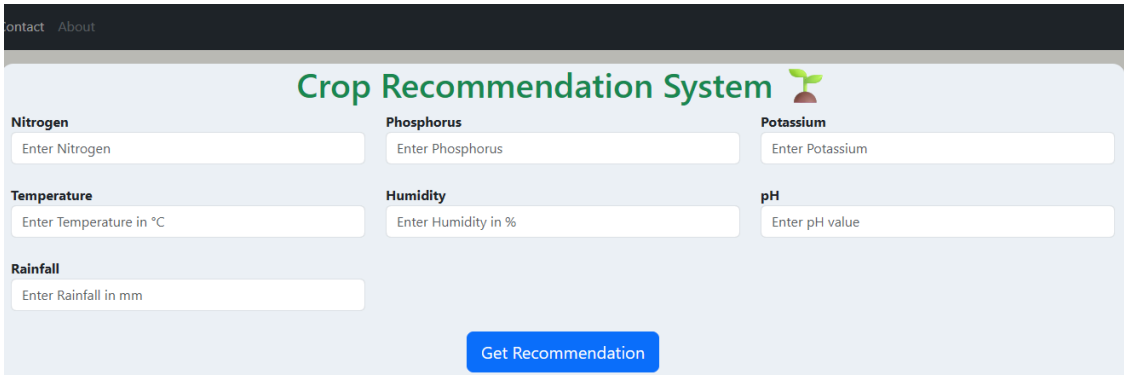
Objectives

- **Automated Location Detection & Environmental Data Collection**
  Develop a GPS-based location detection feature that automatically identifies the user's coordinates and retrieves relevant environmental data from public APIs.
- **Intelligent Analysis & Top 3 Crop Recommendations**
  Build an analysis engine that processes collected environmental data to generate the top 3 crop recommendations best suited to the user's location, each accompanied by a confidence score and clear reasoning.
- **Google Gemini AI Chatbot Integration**
  Integrate the Google Gemini API as an interactive personal agricultural assistant, capable of answering questions and providing guidance on planting techniques, crop care, pest control, and harvesting in natural language.

Review of Existing Solutions

Existing crop recommendation applications have demonstrated the value of data-driven approaches in agriculture, yet several notable limitations persist across current solutions. Three publicly accessible systems illustrate these gaps well.

Noor Saeed's Crop Recommendation System applies multiple machine learning algorithms to soil and climate inputs, but requires users to manually gather and enter all data themselves. It lacks real-time data integration, provides no quantified explanation of how each factor influences the recommendation, and offers no interactive support once a result is returned.



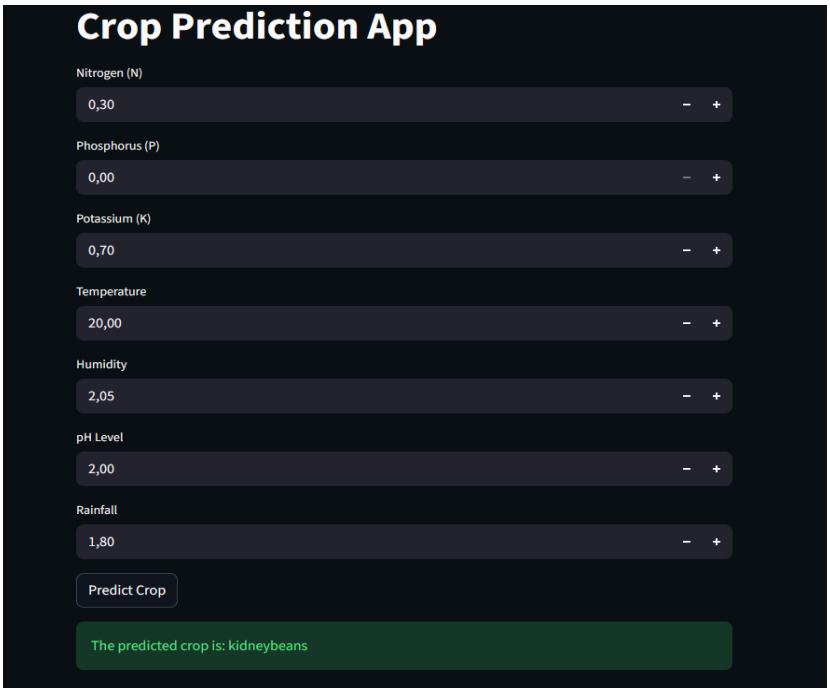
An image of the main page of Noor Saeed's crop recommendation web application.

Jyoti's mobile-based crop recommendation application similarly relies on manual input and has been acknowledged by its own developer to lack quantified effect sizes and key weather variables such as temperature, humidity, and hours of sunlight. Like Noor

Saeed's system, it provides no conversational or advisory layer for farmers to seek further guidance.

An image of the main page of Jyoti's crop recommendation mobile application.

Rakib's Streamlit web application accepts a broader set of weather variables but still depends entirely on manual data entry and outputs only a single crop prediction with no factor-level explanation. It too provides no interactive support beyond the initial result.



An image of the main page of Rakib's Streamlit web application.

Across all three, the same core weaknesses appear: manual data collection burden, directional rather than quantified recommendations, and a complete absence of interactive or advisory support after the prediction is made.

GrowEasy addresses all of these gaps in a streamlined package. Requiring only a GPS location, it automatically retrieves real-time weather and soil data, thus eliminating any manual effort from the farmer. Its machine learning model quantifies each factor's contribution to the prediction, and its Google Gemini-powered AI agronomist provides personalized advisories and an interactive chat interface for follow-up questions, making it a significantly more accessible and intelligent solution than any of the reviewed alternatives.

Methodology

Tech Stack

- **Backend:**
  - GO
  - Gin (Framework)
  - PostgreSQL (Database)
  - GORM (ORM)
  - JWT (Auth)
- **Frontend:**
  - React Framework + Vite
  - JavaScript
  - ESLint (Plugin)
  - CSS (Styling)
- **Machine Learning:**
  - Python
  - FastAPI (API)
  - Scikit Learn, XgBoost, Lgbm (Machine Learning)
  - Optuna (Hyperparameter tuning)
- **External API:**
  - Open-Meteo (Free Open Source Weather API)
  - SoilGrids/ISRIC (World Soil information API)
  - Gemini Flash (Chatbot and AI Summary)
- **Training Dataset:**
  - Crop Recommendation Dataset by Atharva Ingle via Kaggle

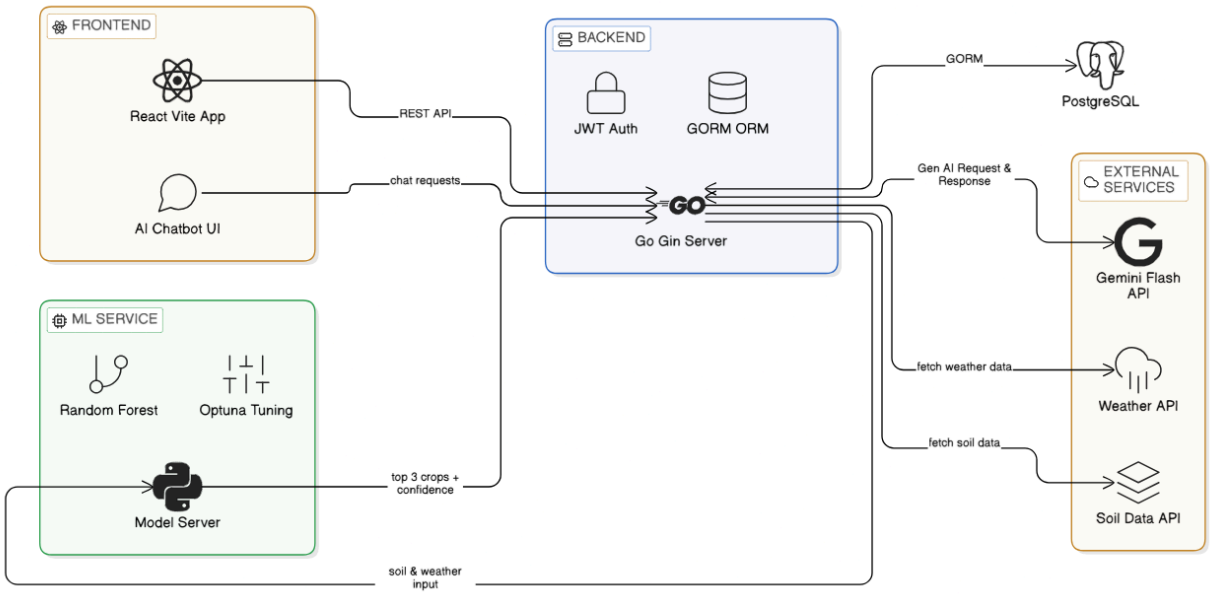
Architecture & Implementation

The system is built on a three-layer architecture consisting of a React-based frontend, a Go backend, and a Python-based ML service, all communicating via REST APIs.

Machine Learning, using a supervised learning approach was used to build the crop recommendation engine. Random Forest was selected as the primary model, with XGBoost and LightGBM used for performance comparison. Optuna was used for hyperparameter tuning to optimize model accuracy. The trained model takes soil and weather features as input and outputs the top 3 recommended crops with their respective confidence scores.

Backend, using Go built with the Gin framework, handles user authentication via JWT and manages data persistence using PostgreSQL through GORM. It serves as the central orchestrator of the system, coordinating communication between the frontend, external APIs, the ML service, and the Gemini Flash API.

Frontend, using React + Vite to provides the user interface for auth, triggering analysis, displaying results, and interacting with the AI chatbot.



How It Works

Analysis Flow

1. The user initiates an analysis from the website
2. The frontend captures the user's GPS coordinates and sends them to the backend
3. The backend requests weather data from **Open-Meteo** and soil data from **SoilGrids/ISRIC** using the received coordinates
4. The collected weather and soil data are forwarded to the **ML model**, which returns the top 3 crop recommendations with confidence scores
5. The backend sends the environmental data and prediction results to **Gemini Flash** to generate a natural language summary
6. The final results are stored in the database and returned to the frontend
7. The frontend displays the analysis results, crop recommendations, confidence scores, and AI-generated summary to the user

Chatbot Flow

1. The user types an agricultural question into the chatbot interface on the website
2. The frontend sends the message along with the conversation history to the backend
3. The backend forwards the message to the **Gemini Flash API**, providing agricultural context to guide the response
4. Gemini Flash processes the query and returns a natural language response
5. The backend stores the chat history in the database and returns the response to the frontend
6. The frontend displays the response to the user, maintaining a continuous conversational experience

Results

Objective	Status
Automated Location Detection & Environmental Data Collection	Success
Intelligent Analysis & Top 3 Crop Recommendations	Success
Google Gemini AI Chatbot Integration	Success

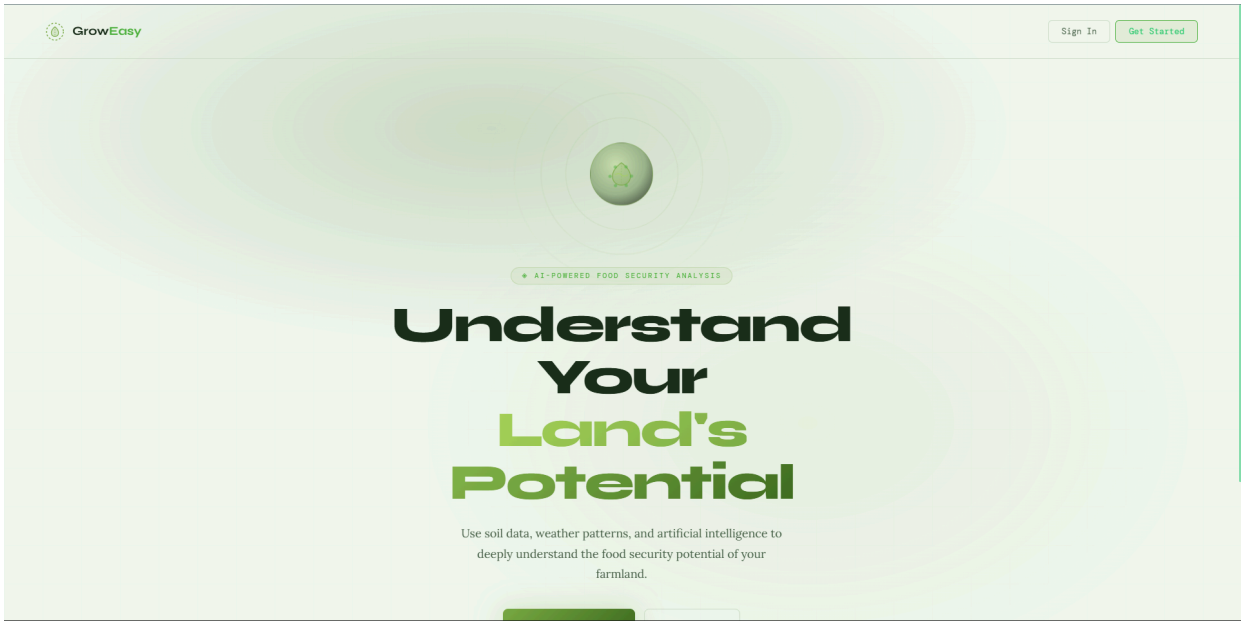
The website successfully detects the user's GPS coordinates, retrieves relevant environmental data, and generates the top 3 crop recommendations with confidence scores and reasoning. The Google Gemini-powered chatbot is fully functional as an interactive agricultural assistant, capable of answering questions on planting techniques, crop care, pest control, and harvesting in natural language.

Limitation and Known Issues

- The external environmental data API used for ML inference experiences occasional downtime, which may prevent the crop recommendation feature from running entirely until the service is restored.
- The Google Gemini Flash API, used for the chatbot and data summarization, may experience slower response times during peak hours due to free-tier usage limits.

Prototype Showcase

Landing Page



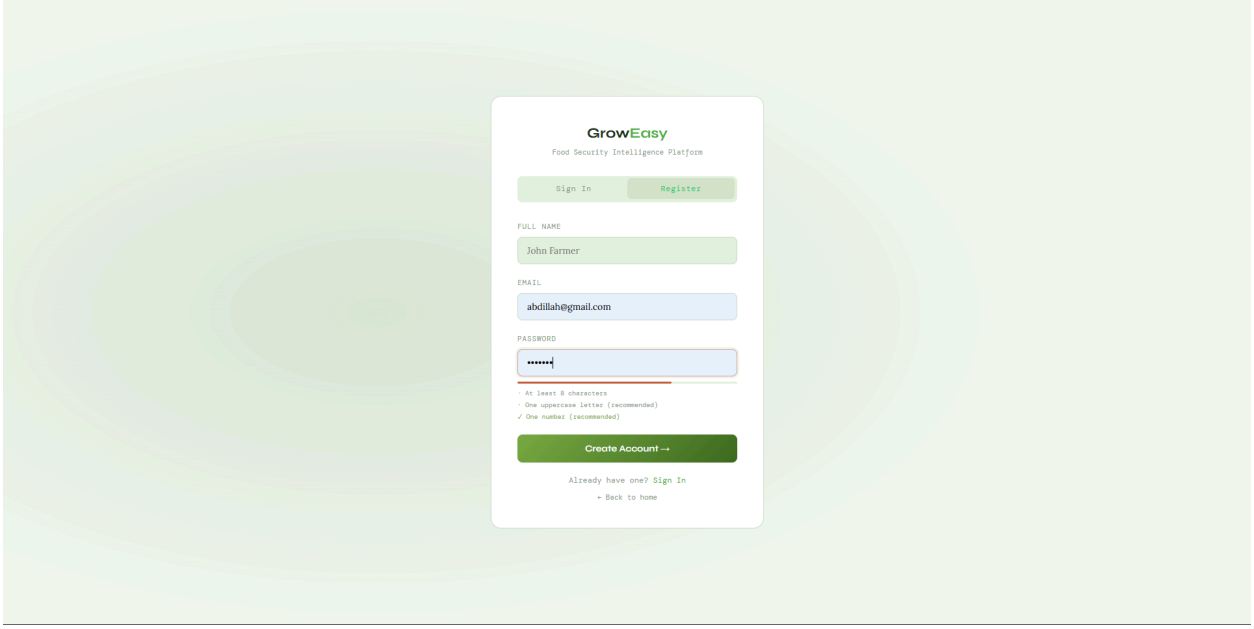
The landing page has an organic-digital aesthetic, a light sage green background with a barely-visible grid overlay and two soft green glowing orbs, making the page feel alive and soft to look at.

Navigation sits at the top with the GrowEasy logo on the left and two buttons on the right: a ghost-colored Sign In button and a green-tinted Get Started button.
The hero section is fully centered and flows top to bottom:

- A floating animated orb with pulsing rings, a rotating orbit, and a leaf icon inside
- A green pill badge labeling the product as AI-powered
- A large headline where "Land's Potential" is rendered as a green gradient text
- A short serif body paragraph explaining the product
- Two CTA buttons — a solid glowing green Start Analyzing → (primary) and a ghost View Demo (secondary).

Typography uses three fonts Syne (bold headings), DM Mono (labels and UI), and Lora (body text) balancing a techy feel with organic warmth. The color palette stays entirely in greens, from bright lime accents down to deep forest tones.

Register Page



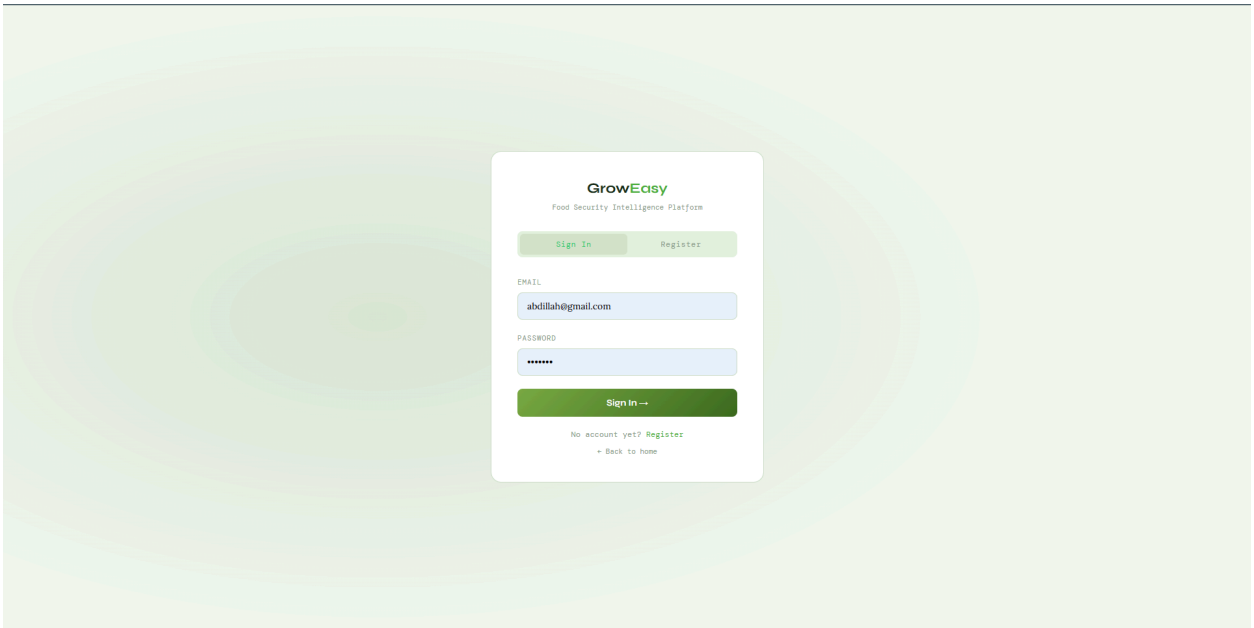
The registration page is designed with a clean, minimalist UI to ensure a seamless onboarding experience for users. It focuses on security and ease of use, acting as the gateway to the Food Security Intelligence Platform.

Key Features:

- Dual-Mode Interface: Users can easily toggle between Sign In and Register using the intuitive tab switcher at the top.
- Secure Data Collection: Collects essential user information, including Full Name, Email, and Password, to create a personalized dashboard.

- Live Password Strength Validator: * Provides real-time feedback on security requirements (e.g., minimum 8 characters, uppercase letters, and numbers).
- Features a visual strength bar indicator to encourage users to create robust, hack-resistant passwords.
- Quick Navigation: Includes "Back to Home" and "Sign In" shortcuts to minimize friction and improve the overall User Experience (UX).

Login Page

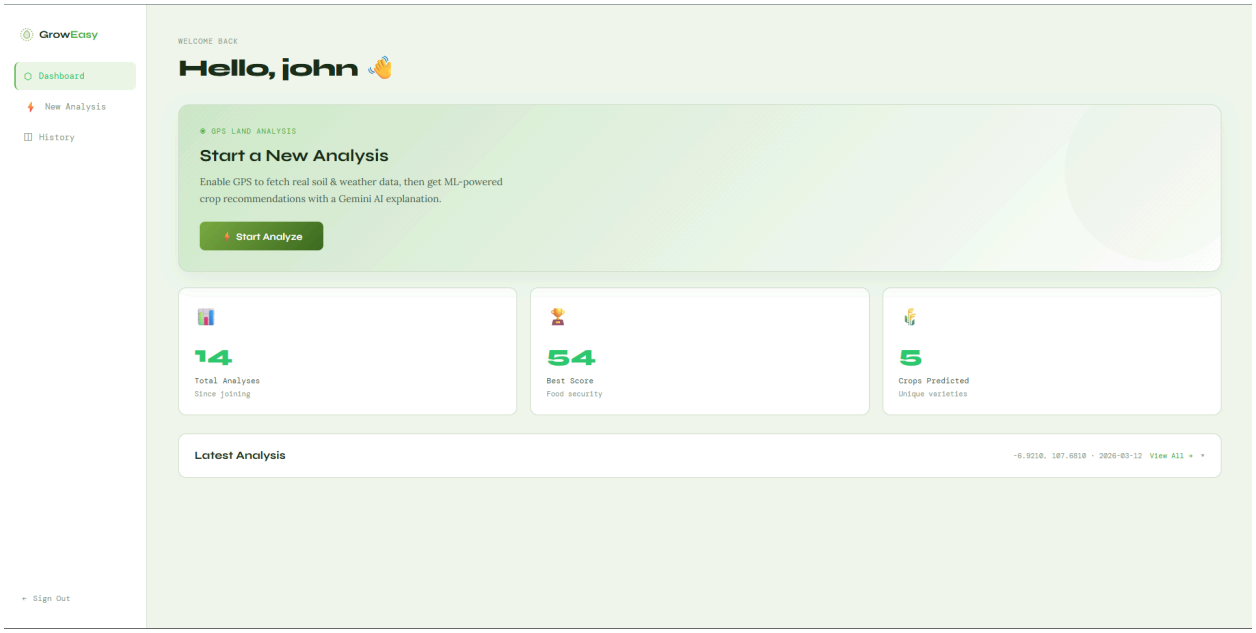


The GrowEasy Sign-In Page provides secure access to our AI-powered agricultural assistant. The design focuses primarily on accessibility and ease of use to empower user productivity.

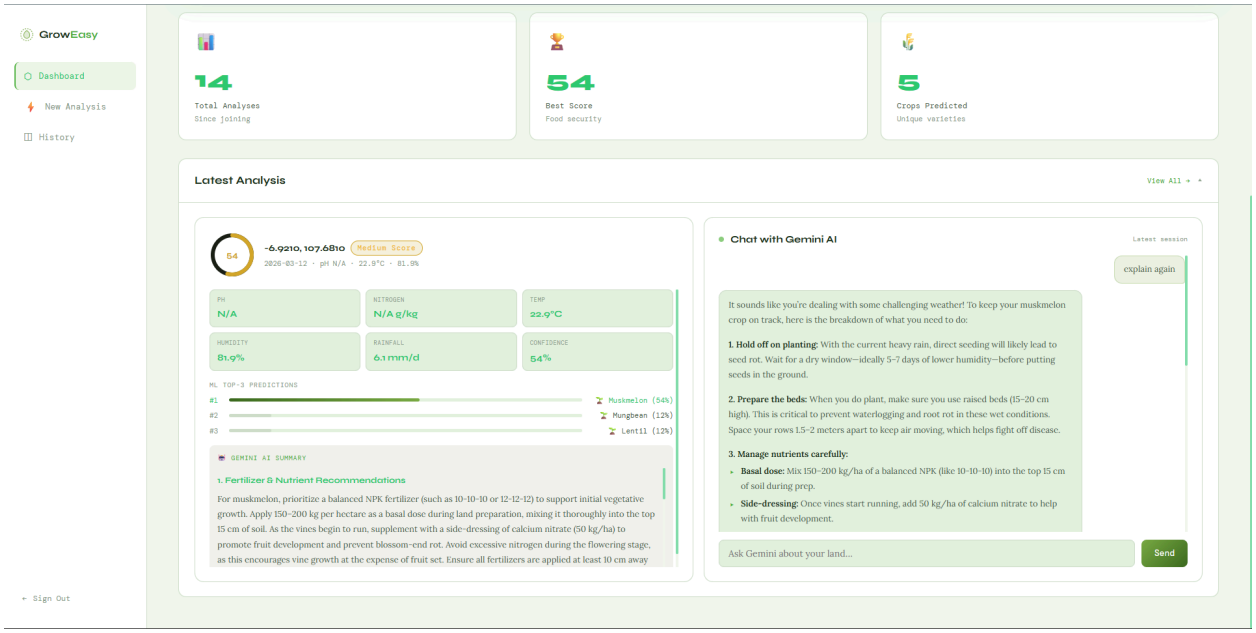
Key Features:

- Intuitive Login Flow: Users can quickly authenticate using their registered Email and Password.
- Segmented Control: Features a top tab switcher that allows users to seamlessly pivot between Sign In and Register views without page reloads.
- Visual Feedback: The input fields utilize clear labels and a consistent color scheme to guide the user throughout the authentication process.
- Call to Action (CTA): A prominent green button provides a clear path for users to enter their personalized dashboard.
- Accessibility Links: Includes a "Register" link for new users and a "Back to home" option to ensure users never feel "stuck" on the login screen.

Dashboard Page

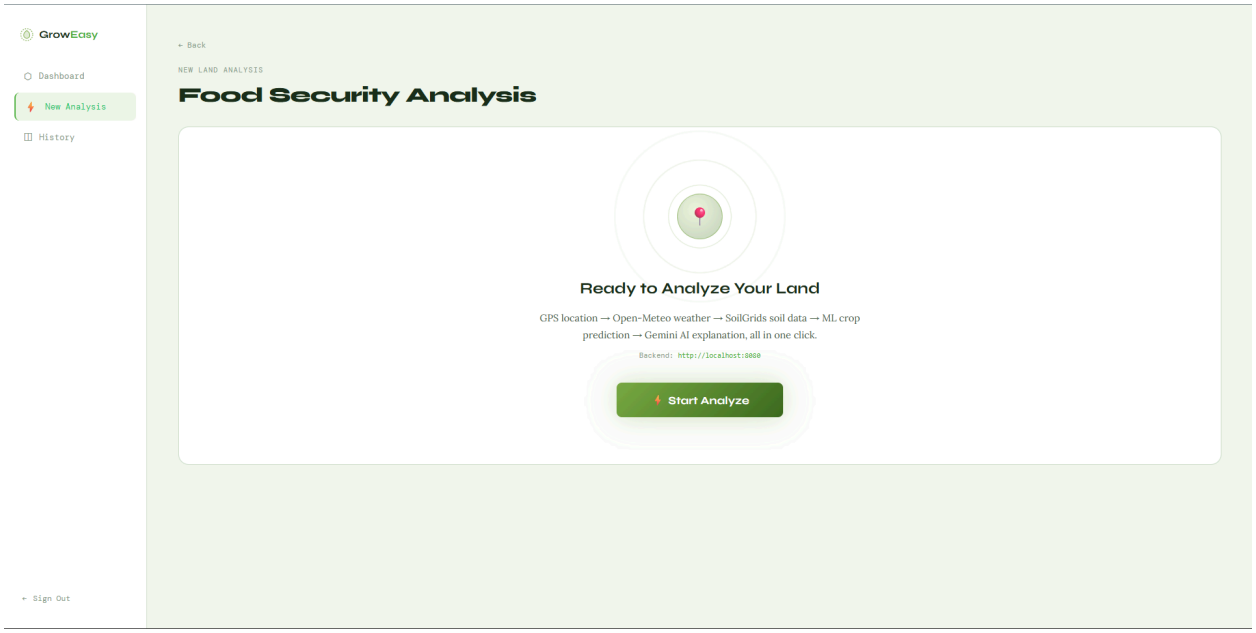


After we authorize the application, first of all, we would see the Dashboard page. Here we can start new analysis by clicking the start Analyze button. Here we can also see the number of analyses that we have done, the highest score we got, and the number of crops that the system predicted for us. We could also see the latest analysis that we have done. If we click the latest analysis bar, we would get the display like below:

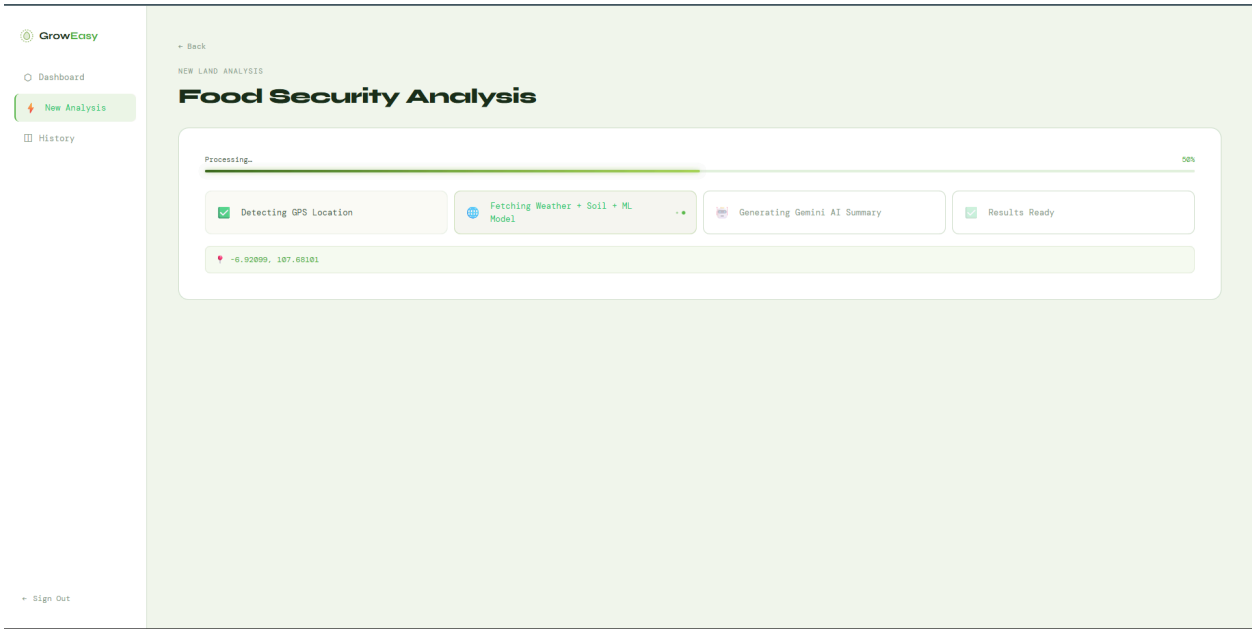


After we clicked the latest analysis, we could see our latest analysis. It contains the longitude and latitude of our location right now, soil data, the weather data, top three crops result, AI Summary, and we could also resume our chat of our latest analysis.

Analysis Page

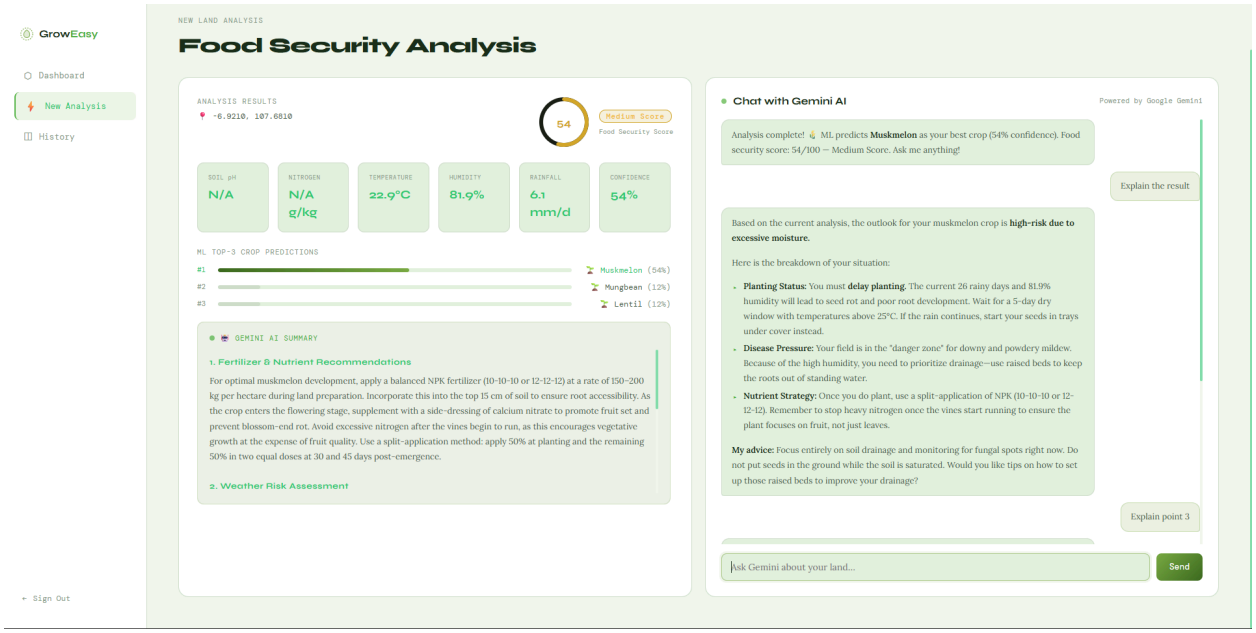


If we click the start analyzing at the dashboard, we would land on the analysis page. We could start the analysis by clicking the Start Analyze button.



After we click the start analyze button, the system will start to analyze (starting from collecting user location, collecting weather and soil data from API then inference with

ML model, and generating AI summary). There is a progress bar so the user can see what happens behind the system and make the user experience better.



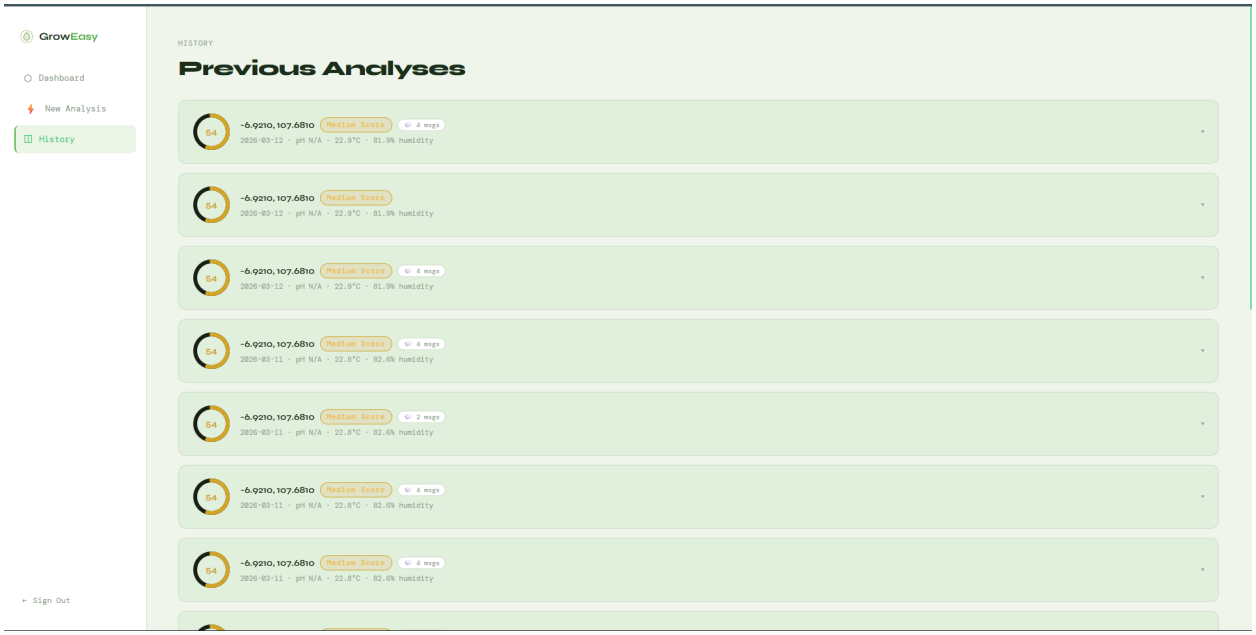
After the analysis is over, we can see the result. We can see the input data such as location (longitude, latitude), nitrogen, ph, temperature, humidity, and rainfall. Rain and ph value is N/A, because when we run the demo the soilgrids api is error (if the API is functioning correctly, it would show the data). Although the API is error prone, we can still use the system based on weather alone, ensuring the durability of the system although there is error in the external API.

We can see the result of the prediction (top three crops) below the input data. As shown in the screenshot the third best is muskmelon followed by mungbeans and then lentil. There is a percentage showing how confident the system is about the prediction (the highest confidence score is also shown as a chart, as you can see there is a chart that displays a yellow circle that has a 54 medium score).

Below the crop prediction we could see the prediction summary generated by gemini. There is information like fertilizer and nutrient recommendation, weather risk assessment, etc.

On the right side we can chat with the chatbot about the crop prediction. We could ask the AI to explain further about the result of the prediction, what the next step that we have to do, and other things that are related to the crop prediction.

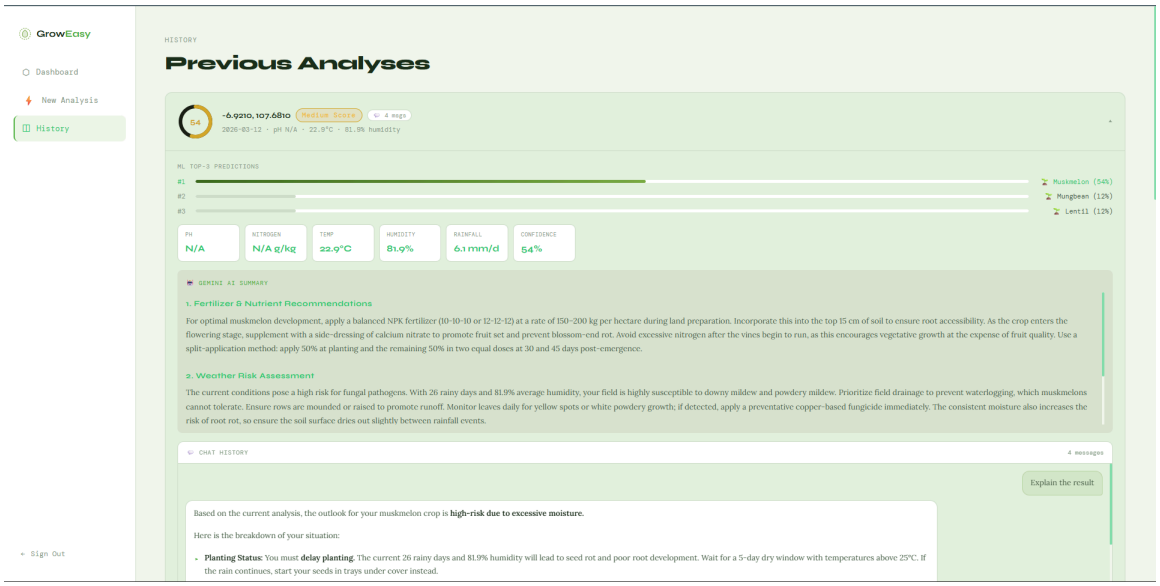
History Page

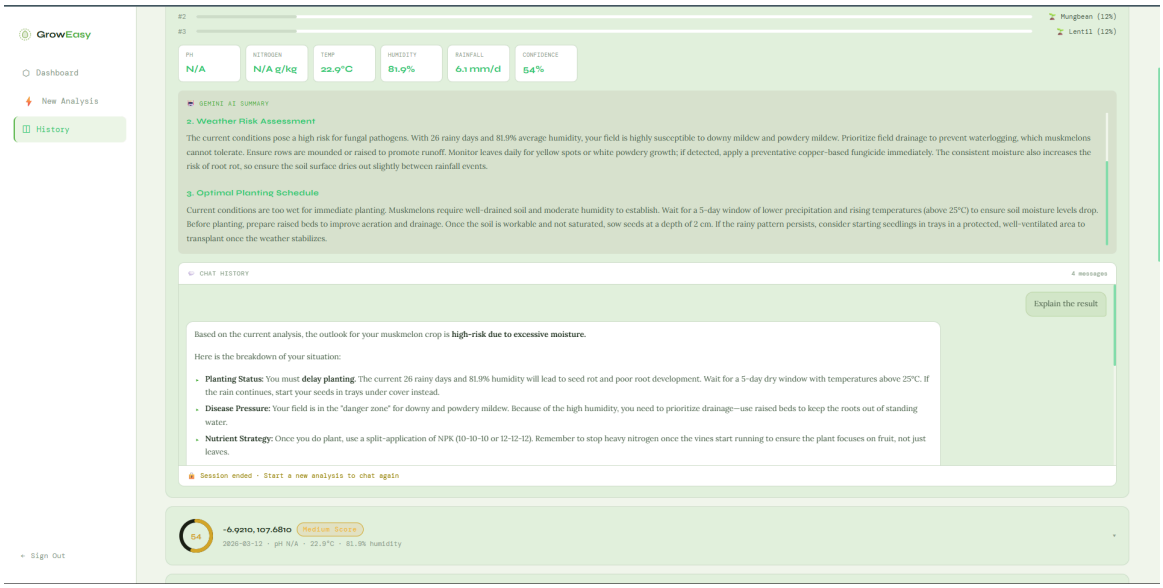


The History Page in the GrowEasy application serves as a centralized hub for analysis records, allowing users to revisit every assessment ever performed on a specific plot of land or location. This page does not merely store raw data; it archives machine learning predictions, AI-generated summaries, and historical consultation logs.

Key Features (Collapsed View):

- Analysis Score: A circular indicator showing the land's overall suitability for farming.
- Location Coordinates: Displays specific Latitude and Longitude for precise tracking.
- Result Category: Instant labels (*Medium Score*) for quick status interpretation.
- AI Chat Indicator: Shows the number of messages exchanged with the AI for additional insights.
- Environmental Metrics: Highlights core parameters: Soil pH, Temperature, and Humidity.
- Timestamp: Records the exact date to monitor land progress and historical trends.





Analysis Details (Dropdown View)

When expanded, each record provides a deep dive into the land's data through four key modules:

A. Machine Learning Predictions

- Top 3 Crop Suitability: Displays the best crops to plant based on environmental parameters (Temp, Humidity, Rainfall, Soil Nutrition, and pH).
- Confidence Ranking: Uses visual progress bars to show the probability of each recommendation (*Muskmelon – 54%*).

B. Environmental Data Transparency

- Input Metrics: Shows the exact data used by the model (Nitrogen, pH, Rainfall, etc.).
- Model Reliability: Includes a Confidence Score to ensure transparency in the AI's decision-making process.

C. AI Agronomy Summary (Generative AI)

- Fertilizer & Nutrients: Tailored soil treatment plans for the predicted crops.
- Risk Assessment: Identifies threats like high humidity or potential crop diseases.
- Optimal Planting Schedule: Strategic timing based on current soil and weather conditions.

D. AI Chat History

- Consultation Logs: Saves all past interactions with the AI assistant.
- Context Retention: Allows users to revisit previous advice or continue discussions without losing context.

AI Acknowledgement

AI Tool	Usage
Claude Sonnet	Used for code generation and debugging code
Claude Sonnet	Used for refining and polishing document wording
Gemini Nano Banana	Used for image generation to make images clearer
Gemini Flash	Used for AI Chatbot and Summary integration
Eraser.io	Used for generating architecture diagram
Google AI Studio	Used for Text-to-Speech in Demo Showcase Video
Capcut AI	Used for Text-to-Speech and Subtitle in Demo Intro Video

Conclusion

GrowEasy successfully delivers on all three of its core objectives: automated GPS-based environmental data collection, intelligent top 3 crop recommendations with confidence scores, and an interactive AI-powered agricultural chatbot. By combining user's environment data with machine learning inference and natural language AI, GrowEasy provides an accessible, location-aware crop recommendation platform that requires no agricultural expertise or manual data input from the user. In doing so, it directly addresses the gaps identified in existing solutions and contributes meaningfully toward SDG 2 (Zero Hunger) and SDG 9 (Industry, Innovation and Infrastructure).

Future Improvements

Several enhancements could further strengthen GrowEasy in future iterations:

- **Alternative & More Reliable External APIs**, Explore and migrate to more reliable external data sources with higher uptime guarantees, such as paid weather and soil data APIs to ensure the crop recommendation feature remains consistently available to users.
- **Crop Calendar & Seasonal Guidance**, Extend recommendations to include optimal planting and harvesting schedules based on seasonal patterns for each recommended crop.
- **Multilingual Support**, Add another language besides english support to make the platform more accessible to the broader audiences.
- **API Tier Upgrade**, Upgrade to a paid Gemini API plan to eliminate response delays during peak hours and ensure a consistently smooth chatbot experience.

References

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